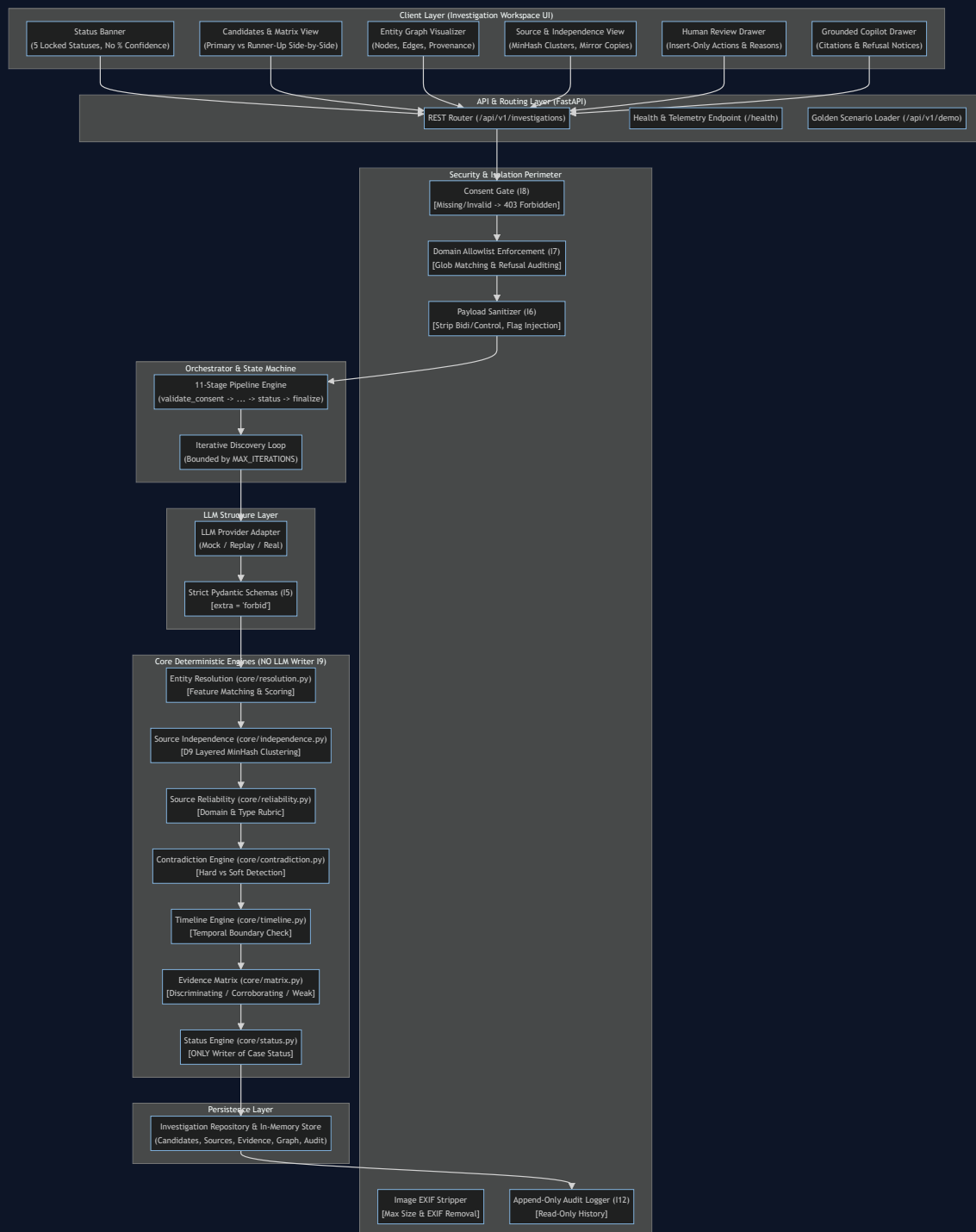


TRACEID — Complete System Architecture & Pipeline Diagrams

This document contains the authoritative architecture diagrams, system component flowcharts, security boundary models, decision rule flowcharts, and data flow figures for the **TRACEID Evidence-Based Digital Identity Intelligence Platform**.

1. High-Level System Component Architecture

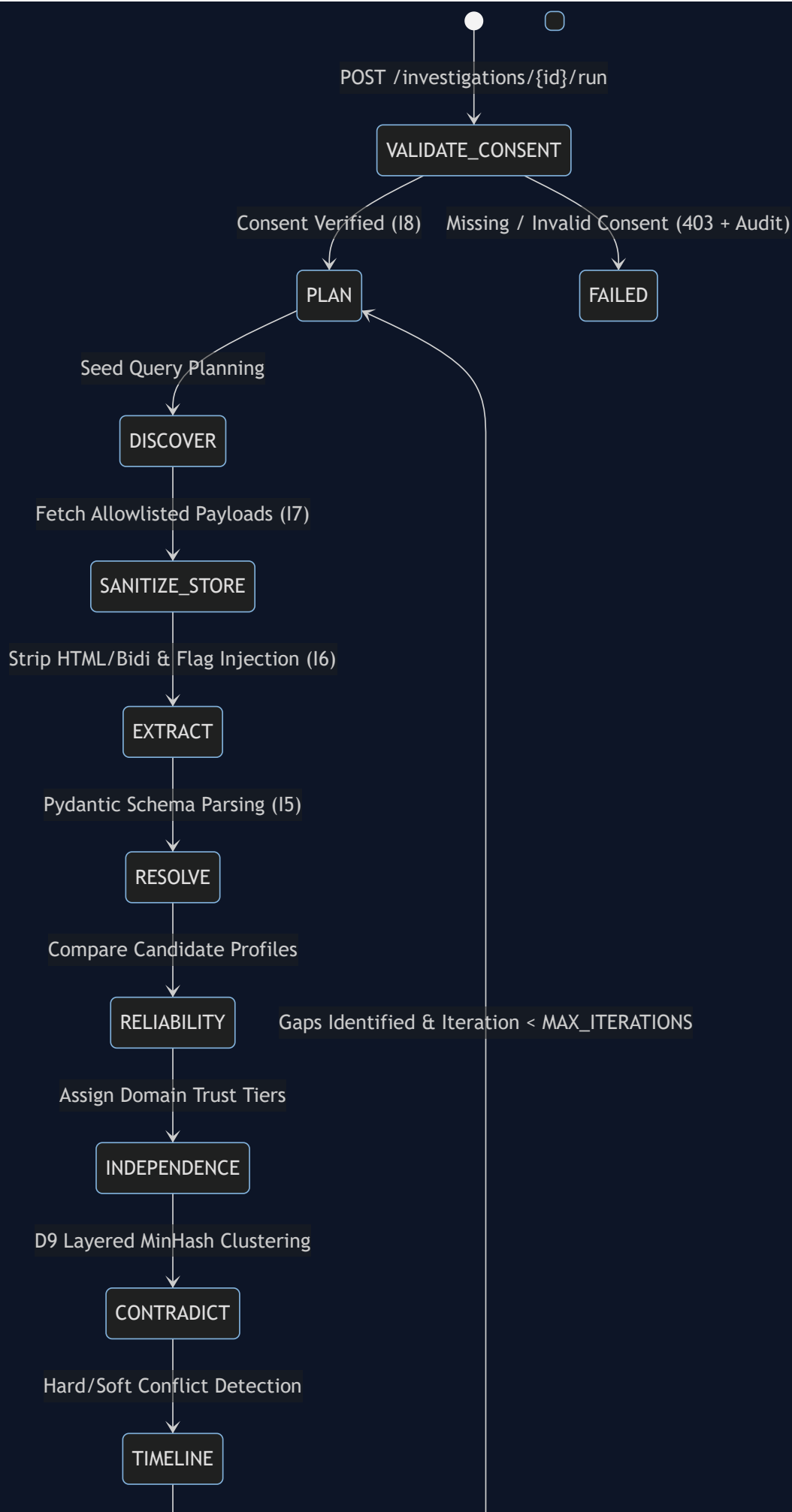
The high-level architecture separates untrusted external data, the security perimeter, deterministic execution engines, LLM structure enforcement, and the interactive cyber-intelligence workspace UI.

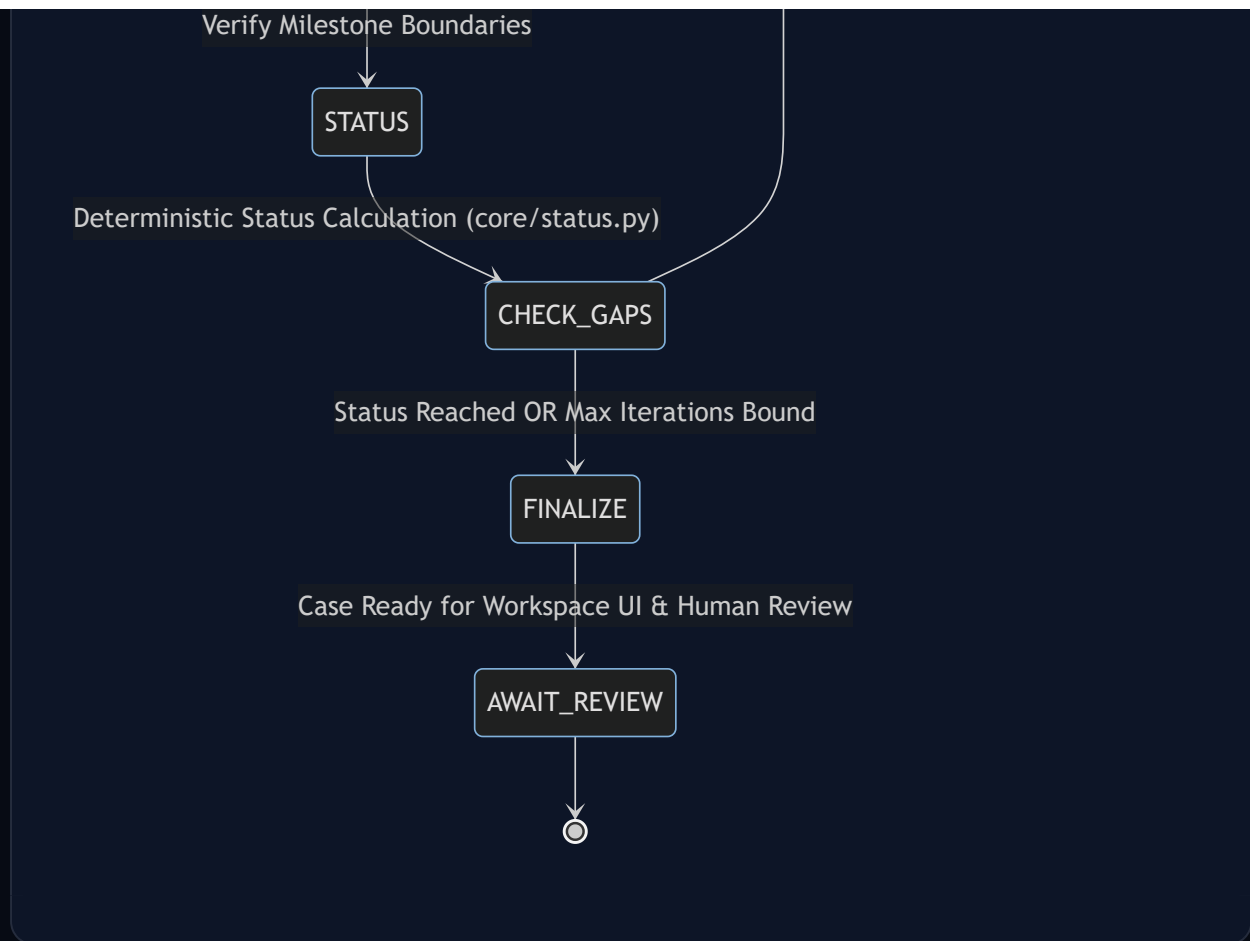


2. Orchestrator Pipeline State Machine (11 Locked Stages)

The investigation pipeline moves sequentially through 11 locked stages. If evidence gaps are discovered, an iterative discovery loop triggers additional allowable fetches up to `MAX_ITERATIONS`.



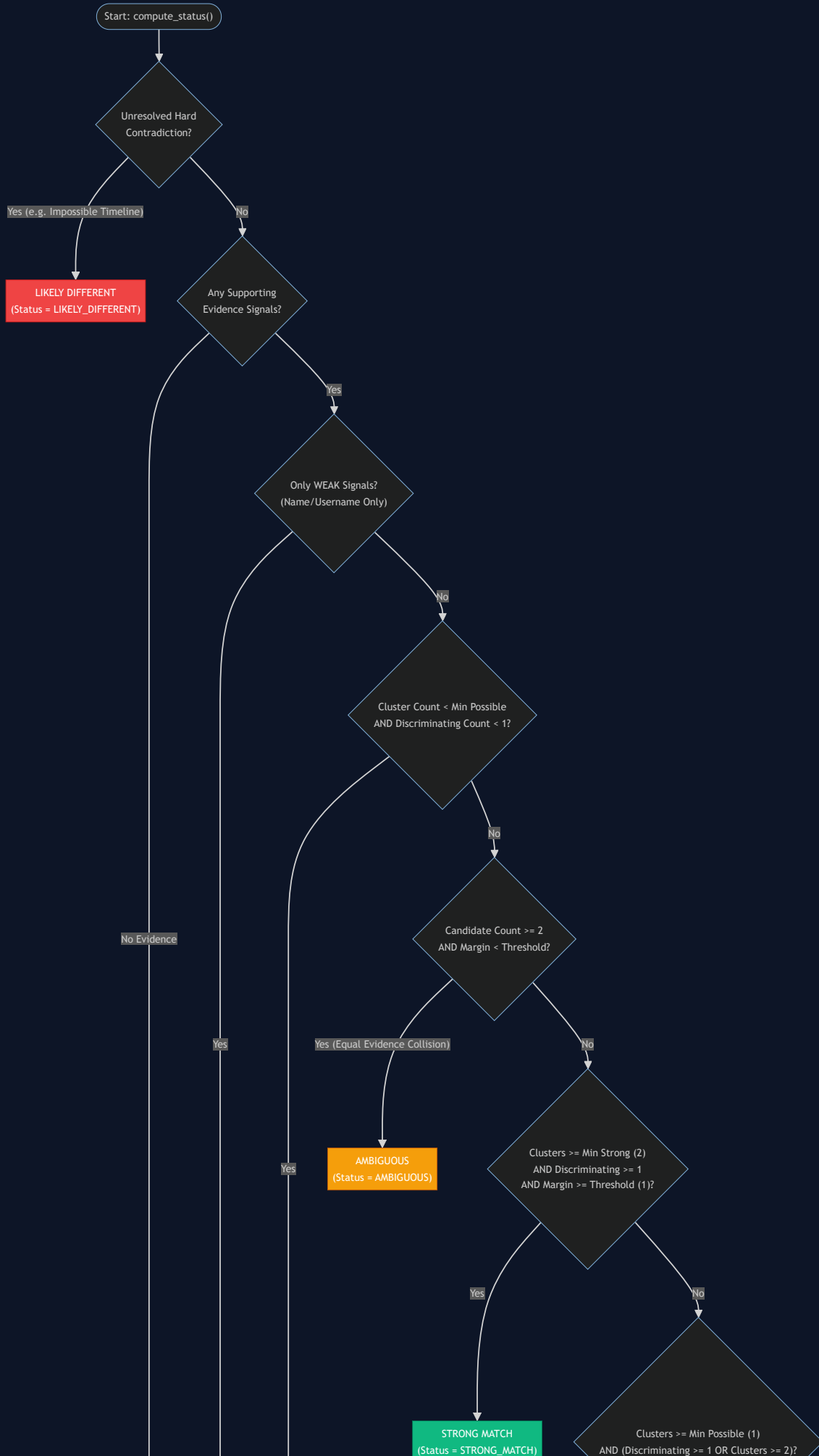




3. Decision Rules v1 — Deterministic Status Engine (`core/status.py`)

Case status is computed **exclusively** by `core/status.py` . The LLM never writes or overrides the case status.

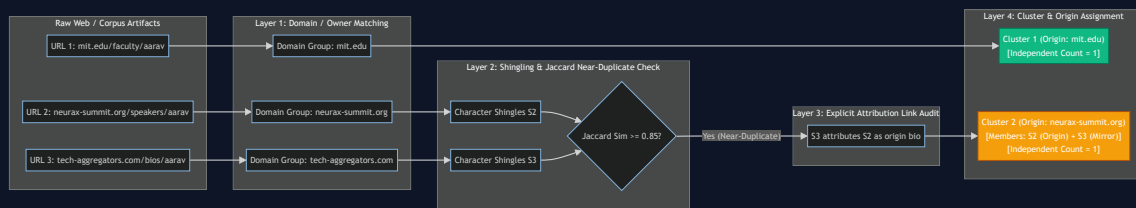






4. Source Independence & MinHash Clustering (D9 Layered Method)

To prevent syndicated news, web mirrors, or social copy-paste from inflating identity match confidence, sources are clustered into independent groups using a 4-layer evaluation.



5. Summary Matrix of Architecture Rules & Guarantees

Rule / Invariant	Enforced Component	Architectural Guarantee
I5 (LLM Schema)	<code>app/llm/schemas.py</code>	All LLM responses parse into strict Pydantic schemas (<code>extra="forbid"</code>). LLM cannot return freeform JSON.
I6 (Sanitizer)	<code>app/security/sanitize.py</code>	Web content treated as DATA, never instructions. Bidi/control characters stripped, injection patterns flagged.
I7 (Allowlist)	<code>app/security/allowlist.py</code>	Glob pattern matching on domain allowlists. Unallowed domains trigger audited refusal responses.
I8 (Consent Gate)	<code>app/security/consent.py</code>	Mandatory consent check before pipeline run. Missing consent returns <code>403 Forbidden</code> + audit log.
I9 (Status Writer)	<code>app/core/status.py</code>	THE ONLY PLACE case status is written. Deterministic, auditable decision rules v1. LLM never writes status.
I10 (No Percentages)	<code>app/core/matrix.py</code>	No single confidence percentages anywhere in backend or UI. Signal tiers (Discriminating/Corroborating/Weak) displayed.
I11 (Review Insert-Only)	<code>app/orchestrator/engine.py</code>	Human review actions (Confirm/Reject with mandatory reason) are append-only. Evidence is never deleted.
I12 (Append Audit Log)	<code>app/security/audit.py</code>	Audit log entries are read-only and append-only.
I13 (Runner-Up Visible)	<code>app/api/router.py</code>	Candidates response always returns runner-up candidate when ≥ 2

Rule / Invariant	Enforced Component	Architectural Guarantee
		candidates exist.
D9 (MinHash Clusters)	<code>app/core/independence.py</code>	Near-duplicate web content and syndicate mirrors grouped into single clusters. Decisions count clusters, not URLs.